In-Class Discussion

EN 601.124: The Ethics of Artificial Intelligence and Automation

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Ravichander et al. categorize hallucinations into Type A (correct fact in training, model fails to recollect), Type B (incorrect fact in training), and Type C (fabrication/over-generalization)

Which of these "error types" is most dangerous for a user to encounter? If a model makes a Type B error because it was trained on "conspiracy theory websites," does the responsibility for that hallucination lie with the model architecture or the data curators?

The authors note that models often hallucinate in scenarios where they should simply abstain (refusal-based tasks). Why? What scenarios are these and which should be included? Why might it be difficult for models to say “no” and abstain?

In class exercise: Select one paper that you liked and provide it as input to HopGPT and NotebookLM. Prompt it with text to feel the robustness to model hallucination and compare the responses.

For example:
Paper selected: Retrieval-Augmented Generation (RAG)
Prompt: Why did the authors of the paper suggest that RAG would solve the political deepfake problem?

There are several methods listed for fact checking against hallucinations: entailment verifier, FactScore, Refusal verifier, python package index, program. Choose one and explain how this works. How could it be embedded within a neural network architecture to counter hallucinations? What are the benefits and cons to this?

Lewis et al.: Retrieval-Augmented Generation (RAG)

RAG combines "parametric memory" (the pre-trained model) with "non-parametric memory" (a dense vector index of Wikipedia). How does the ability to "hot-swap" an index (e.g., replacing 2016 news with 2018 news) change the "accountability gap" described by Raji et al.? Does it make the model easier or harder to audit?

RAG is presented as a way to reduce hallucinations by grounding the model in external facts.

If Ravichander et al. argue that hallucinations are "riddled" even in the best models, can a technical fix like RAG ever truly solve a Type C "fabrication" error? Or is RAG simply a better operationalization of a search engine?

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